Lecture 8:

Model Validation

OUTLINE

8.1 Insight and Illumination

8.2 Validation: When Models Go Bad

8.3 The Techniques of Validation

8.4 Model Discrimination

8.5 Meta-Models

8.6 Précis on Validation

8.1 Insight and Illumination

- Modeling, like computing and statistics, should produce ***insight***, not merely numbers.
- Three general areas that help evaluate the meaning of the number:
  - ***Validity***: Validation concerns the degree of our faith in the quality of the model with respect to the external world.
  - ***Uncertainty***: Ignorance and uncertainty occur at many points in the modeling process: in the equations, the parameters, and in the definition of the system itself.
  - ***Behavior***: The change of state variables over time is the lowest level of system understanding. To grasp fundamental interactions, we need to visualize the covariation between coupled variables, and identify system conditions in which the dynamics of the variables are qualitatively similar.

8.2 Validation: When Models Go Bad

- The system scientists who use the word ***validation*** use it to mean model quality with respect to the objectives of the modeling project.
- More recently, however, several authors have argued for using *corroboration* or *confirmation* for validation.
- The author thinks the adjective *plausible* more accurately reflects the nature of tested biological models and the skeptical attitude we should adopt.
- Model quality, if it is quantifiable at all, is a continuous variable and perfection is probably not achievable.
- The process of model evaluation is unending.

8.2 Validation: When Models Go Bad

- Important Criteria for Quality
  - Usefulness for system control or management
  - Understanding or insight provided
  - Accuracy of predictions
  - Simplicity or elegance
  - Generality (number of systems subsumed by the model)
  - Robustness (insensitivity to assumptions)
  - Low cost of running or constructing the model.
- The model objectives will determine the weighting to be given to the different components.

8.2 Validation: When Models Go Bad

- The Logic of Falsifying Complex Simulation Models
  - An *Aristotelian syllogism* is a sequence of logical steps that in totality is true regardless of the truth or falsity of the component steps.
  - The basis of the modern concept of scientific falsification is a syllogism called *modus tollens*:

Form:	Example:	
$A \Rightarrow B$	*if Spock is human, then he will act illogically.*	(8.1)
$\neg B$	*Spock does not act illogically.*	

$\neg A$	*Therefore, Spock is not human.*	

8.2 Validation: When Models Go Bad

- The Logic of Falsifying Complex Simulation Models
  - In applications of this argument in science, "A" is the general hypothesis (law) and "B" is the implication or prediction that follows from the law in a particular instance.
  - The fallacy of *affirming the consequent*:

Form:	Example:	
$A \Rightarrow B$	*if Frodo loses the ring, then he will be ill.*	(8.2)
B	*Frodo is ill.*	

A	*Therefore, Frodo has lost the ring.*	

8.2 Validation: When Models Go Bad

- The Logic of Falsifying Complex Simulation Models
  - Even though one observes many instances of the major premise, this neither establishes it as a law nor permits one to infer the conditional (A) based solely on the observation of the prediction (B).
  - *Modus tollens* is difficult to implement in mathematical models because the law ("A" in Eq. 8.1) is actually a conjunction of a large number of separate assumptions.

$$(a_1 \wedge a_2 \wedge a_3 \cdots \wedge a_n) \Rightarrow B$$
$$\neg B$$
$$----$$
$$\neg (a_1 \wedge a_2 \wedge a_3 \cdots \wedge a_n)$$

  - We can perform independent experiments to estimate parameters, perform parameter sensitivity analysis to evaluate their effects on model response, or create and investigate alternative models.

8.2 Validation: When Models Go Bad

- The Geometry of Validation
  - Mankin et al. (1977) considered validation in terms of the relation of sets of measurements that can be made on systems and models (Fig. 8.1).

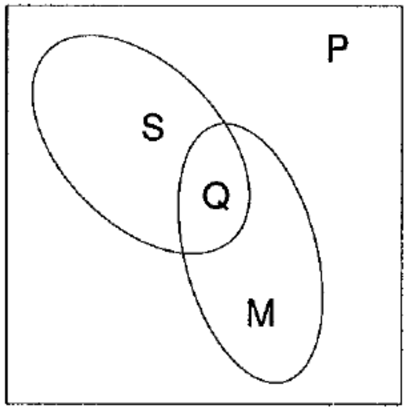


Figure 8.1: Relations of sets of observations on the system (S) and model (M) for validation. Q is the set of correct predictions. (From Mankin et al. (1977), Fig. 1.© 1977 Simulation Councils, Inc. Reprinted with permission Simulation Councils, Inc., publisher.)

8.2 Validation: When Models Go Bad

- The Geometry of Validation
  - There are several qualitative relations between these sets that help us understand different validation situations and ways that models can fail (Fig. 8.2).

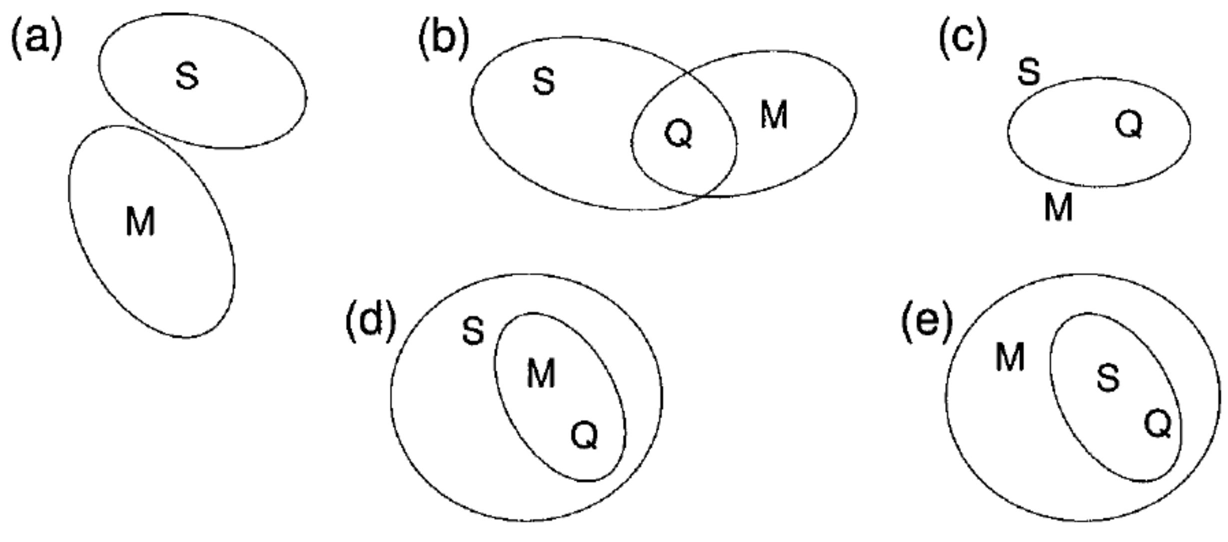


Figure 8.2: Relations of model predictions and system observations.

8.2 Validation: When Models Go Bad

The Geometry of Validation

- If **Q** is empty, there is no intersection between model and observation, and the model is *useless*. If **Q** is nonempty, we say the model is *useful*.
- Mankin et al. (1977) also suggested that *model reliability* is the ratio of the size of **Q** to the size of **M**. *Model adequacy* is the ratio of the size of **Q** to the size of **S**.
- Practically, we can only compare model predictions to observations we have made. Most observational data sets consist of a relatively small number of observations separated by relatively large time or space distances.

8.2 VALIDATION: WHEN MODELS GO BAD

- The Geometry of Validation
  - We must investigate both reliability and adequacy. Most published validation exercises focus on the size of **Q** or, at best, on model adequacy. Model reliability is inherently more difficult to evaluate.
  - *Error analysis* and *sensitivity analysis* can provide insight into the true range of model predictions and, consequently, the size of **M.**
  - To address model reliability, the model must be tested in imaginative ways. For example, it should be tested against
    - Different systems [e.g., different organisms, or habitats (aquatic vs terrestrial)];
    - Different geographical areas;
    - Using different parameter values and environmental driving variables and perturbations.

8.2 Validation: When Models Go Bad

- Variables and Levels for Validation
  - Usually, the model objectives will dictate which quantities should be compared between model and data. The most common are the dynamics of the state variables and *derived measures* in the form of Forrester auxiliary variables.
  - The *derived measures* may be (1) functions of individual state variables [e.g., a state variable scaled to other units], (2) the time or spatial averages or frequency distributions of a state variable, (3) the maximum of a state variable, or (4) the time that a state variable achieves a particular value (e.g., its maximum).
  - We can also use auxiliary variables that are computed from two or more state variables, or ratios of state variables.

8.2 Validation: When Models Go Bad

- Variables and Levels for Validation
  - In addition to choices of variables, there are degrees of comparisons.
  - We seek objective, statistical criteria to compare models and data.

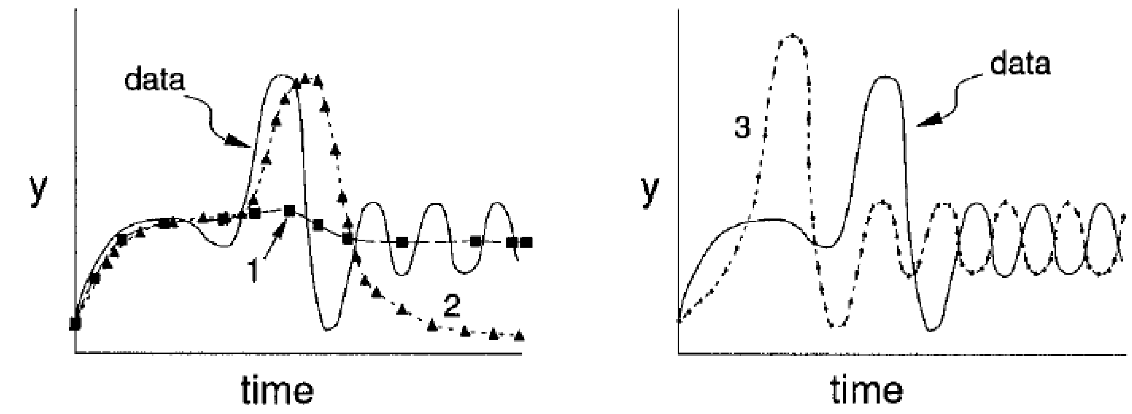


Figure 8.3: Comparisons between data (solid line) and the predictions of three hypothetical models.

8.2 Validation: When Models Go Bad

- Conditions for Validation
  - Four attributes that will influence the kind of validation that is possible:
    - Data Independence
    - Single and Multiple responses
    - Single and Multiple Comparison Points
    - Unreplicated Systems and Models

8.2 Validation: When Models Go Bad

- Conditions for Validation
  - Data Independence
    - The data used for model validation must be separate from and independent of any data used to formulate model hypotheses and estimate parameters.
    - If the comparison data are not independent of those used to construct the model, then we are only doing *calibration* and not true validation.
    - Re-sampling techniques: *Cross-validation*, *jackknifing*, and *bootstrapping* are examples to this approach to pseudo-validation.

8.2 Validation: When Models Go Bad

- Conditions for Validation
  - Single and Multiple Responses
    - Our validation test procedure must decide how many and which of all possible responses will be used.
    - If we choose to validate using more than one response, then we must decide if we will compare system and model for each response *separately* or produce a synthetic validation that incorporates all responses *simultaneously*.
    - Multivariate statistical techniques can perform comparisons simultaneously.

8.2 Validation: When Models Go Bad

- Conditions for Validation
  - Single and Multiple Comparison Points
    - We can choose to validate the model either at a single point in time or at several points in time over the series.
    - If we choose to evaluate the model at a particular point in time, then we must have a criterion for determining what the point will be.
    - If multiple time points are used, then we must use care in applying standard statistical tests.

8.2 Validation: When Models Go Bad

- Conditions for Validation
  - Unreplicated Systems and Models
    - Model validation using statistical tests requires some form of variability in either model predictions or observations.
    - In real systems, variability is usually produced from replicated observations. It can be produced in stochastic models from repeated runs that differ in the sequence of random numbers used to generate the modeled randomness.
    - Regression techniques are one approach to validation that does not require statistical replication.

8.3 The Techniques of Validation

Table 8.1: Summary of quantitative validation techniques. RSS = residual sum of squares; CI = confidence interval.

Method	Definition
1:1 Regression	Regresses data values against model predictions. Simultaneously tests slope = 1.0 and intercept = 0.0. Can be used with or without system replication; ignores temporal sequences. *Page 153*
ρ R^2	Data-model correlation and coefficient of determination. Tests for no correlation. Ignores temporal sequences; does not require replication. *Page 153*
Lack-of-Fit	Tests if the data-model relationship is linear. Requires replicate system observations at each model prediction point. Ignores temporal sequences. *Page 154*
paired t-test	Tests that data-model pairs are equal. Ignores temporal sequence; does not require system replication. *Page 154*
Profile	Tests multiple model variables simultaneously for parallelness with data over time; requires system replication. *Page 159*
95% CI	Semi-quantitative; identifies time values when model or data 95% CI do not overlap with data or model predictions. Requires either system replication or stochastic model predictions. *Page 163*
Turing	Qualitative test using a human expert. May use any system trait or variable; often uses temporal sequences; does not require system replication. *Page 151*

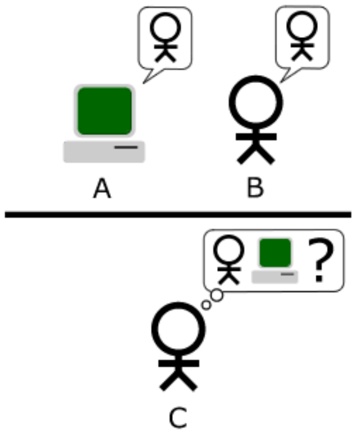
8.3 The Techniques of Validation

LRT	LRT = Likelihood Ratio Test: Compares a set of models by testing that the ratio of likelihoods of a simple model to the best model equals 1.0. Requires nested models. *Page 164*
AIC	Index of model quality based on log-likelihood of model in which quality decreases with model complexity. Not a statistical test. *Page 169*
Bayes	Combines likelihood measures of model error with model prior probability to compute the posterior probability that the model is true relative to a set of models. Not a statistical test. *Page 172*
MSEP RMSEP	Index of model quality: MSEP = mean squared error of predictions [(observed-predicted)/n, units as square of variable units, e.g. [gm C]^2]. RMSEP = square root of MSEP (same units as variable). Error partitioned among: bias, slope, and random. *Page 157*
MAE MA%E	Index of model quality. Absolute value of difference between data and model, same units as data variable. *Page 157*
EF	Index of model quality. EF (Model Efficiency) $= 1 - (\mathrm{RSS}/\sum(y_i - \bar{y})^2)$. Model error scaled to data variability; unitless. *Page 158*
Theil's U	Index of model quality. Model error scaled to variability in model and data. *Page 156*
Janus (J^2)	Index of model quality. Ratio of error using independent data to error using calibration data: $J^2 = \mathrm{MSEP}_{\mathrm{val}}/\mathrm{MSEP}_{\mathrm{cal}}$. *Page 158*

8.3 The Techniques of Validation

Unreplicated Systems – Turing Tests

- The *Turing test* is a proposal for a test of a machine's ability to demonstrate intelligence.
- The "standard interpretation" of the Turing Test, in which player C, the interrogator, is tasked with trying to determine which player - A or B - is a computer and which is a human. The interrogator is limited to using the responses to written questions in order to make the determination.
- This approach can be used for biological models by asking experts to distinguish similarly prepared figures or reports of genuine and simulated system dynamics.



8.3 THE TECHNIQUES OF VALIDATION

- Unreplicated Systems – Observed vs Predicted Regression
  - Methods for model validation:
    - Scatter Plots
    - The Correlation Coefficient
    - 1:1 Regression
    - Test for Lack-of-Fit
    - Paired t-test

8.3 The Techniques of Validation

- Unreplicated Systems – Observed vs Predicted Regression
  - Scatter plots:

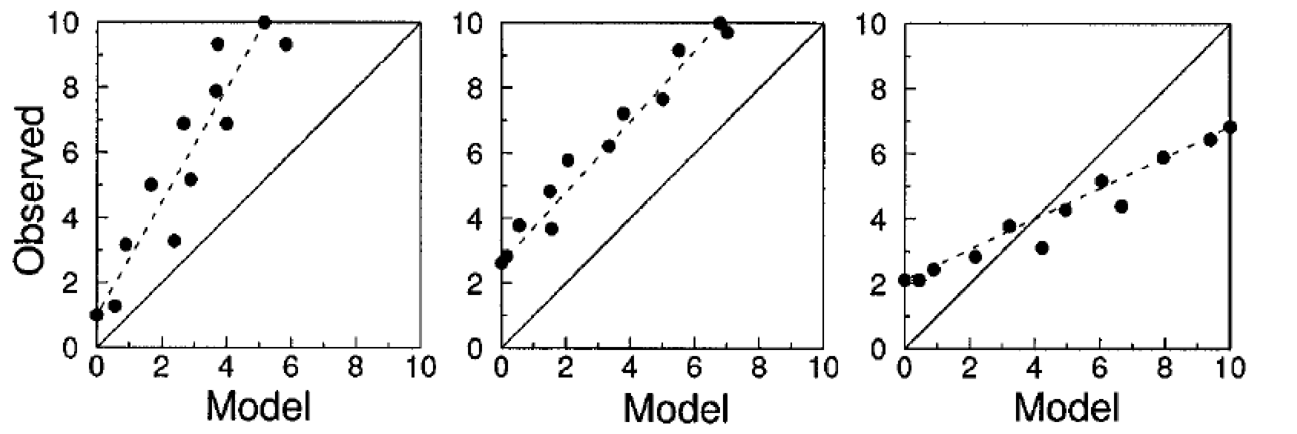


Figure 8.4: Three possible scenarios in which a model with poor fit to data results in a high correlation. Solid circles are data-model pairs and the dashed line is the regression of observations on model predictions. The solid line is the 1:1 line for a perfect fit between data and model. Left: slope (variance) error; Middle: bias error; Right: bias and slope error.

8.3 The Techniques of Validation

- Unreplicated Systems – Observed vs Predicted Regression
  - Correlation coefficients:
    - The correlation coefficient, *r*, measures the strength of the straight-line relation between model and data.
    - While statistical analyses exist to test *r* = 0 (no correlation), there are no *a priori* non-zero values of *r* against which to test.
    - Sample Pearson correlation coefficient

$$r = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}} = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{X_i - \bar{X}}{s_X}\right)\left(\frac{Y_i - \bar{Y}}{s_Y}\right)$$

8.3 The Techniques of Validation

- Unreplicated Systems – Observed vs Predicted Regression
  - 1:1 Regression:
    - The F-statistic to test the slope = 1.0 and intercept = 0.0, simultaneously:

$$F = \frac{na^2 + 2a(b-1)\sum X_i + (b-1)^2 \sum X_i^2}{2s_{Y\cdot X}^2} \qquad (8.2)$$

$$s_{Y\cdot X}^2 = \frac{\sum (Y_i - \hat{Y}_i)^2}{n-2}$$

$$\hat{Y}_i = \overline{Y} + b(X_i - \overline{X}) = a + bX_i$$

Small values of F mean the model is a good fit. This statistic follows the F distribution with 2 and n-2 degree of freedom.

8.3 The Techniques of Validation

- Unreplicated Systems – Observed vs Predicted Regression
  - Test for Lack-of-Fit:
    - This statistic measures the degree that the model does not fit the observations.
    - The model is validated if we do not reject the null hypothesis.
    - This method requires replicated observations at every model prediction used in the test
    - Error = Pure Error + Lack of Fit
    - Detailed in the paper by Kleinbaum and Kupper (1978)

8.3 The Techniques of Validation

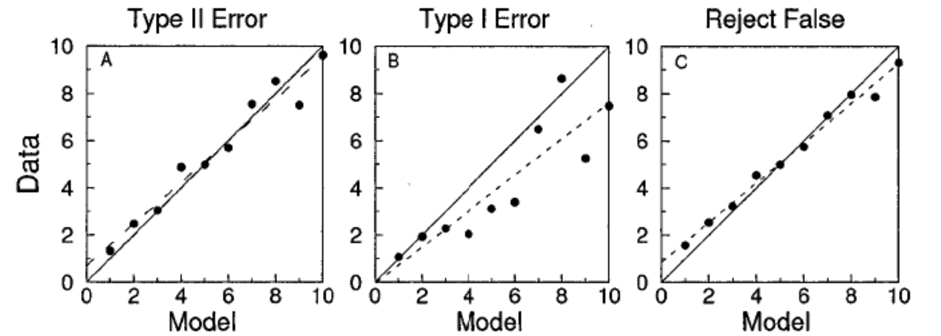
- Unreplicated Systems – Observed vs Predicted Regression
  - Paired t-test:
    - An alternative to 1: 1 regression testing, is to treat model and data as paired samples and use a paired t-test to test H_o : $\mu_X - \mu_Y = 0$.
    - In a comparative analysis, Mayer and Butler (1993) found that 1: 1 regression was more discriminating than a paired t-test; that is, models were rejected using 1: 1 regression, but were accepted in the t-test.

8.3 The Techniques of Validation

- Unreplicated Systems – Observed vs Predicted Regression
  - Type I Error and Type II Error:
    - Type II error: *failure to reject a false null hypothesis*.
    - Type I error occurs when we *reject a true null hypothesis.*
    - We can err if we rely on a single test or index.
    - In large part, choosing which measure to rely on depends on the relative importance one gives to Type I or Type II errors.

8.3 The Techniques of Validation

- Unreplicated Systems – Observed vs Predicted Regression
  - Type I Error & Type II Error:



True: $y = 1.0 + 0.8x$	True: $y = 0.0 + 1.0x$	True: $y = 1.0 + 0.8x$
Estimated: $y = 0.692 + 0.884x$	Estimated: $y = -0.0435 + 0.766x$	Estimated: $y = 1.029 + 0.804x$
Regression: $F = 1.52$ $P > 0.50$	Regression: $F = 6.571$ $P < 0.05$	Regression: $F = 12.52$ $P < 0.01$
Paired $t = 0.222$ $P > 0.50$	Paired $t = -1.832$ $P > 0.10$	Paired $t = -0.2021$ $P > 0.50$
Correlation: $r = 0.978$	Correlation: $r = 0..884$	Correlation: $r = 0.991$
$U = 0.052$	$U = 0.171$	$U = 0.054$
MSEP= 0.412	MSEP= 3.578	MSEP= 0.4239
MC = 0.008 SC = 0.268	MC= 0.495 SC = 0.126	MC = 0.006 SC = 0.752
RC = 0.725	RC= 0.378	RC = 0.242

Figure 8.5: Statistical results of regressing observations on model predictions for three cases with high correlations. Solid circles are model-data pairs; solid line is the 1:1 line; dashed line is the least squares regression line. **A**: Visually, the model looks good, but in reality the model does not match the data. It underpredicts small values and overpredicts large values, yet both the 1:1 regression test and the paired t test fail to detect this. **B**: Visually, the model appears poor. In fact, the model matches the data, but statistical sampling error causes the regression results to indicate a poor model. The paired sample t-test correctly fails to reject the null hypothesis. **C**: A case when the null hypothesis is correctly rejected by F but not t.

8.3 The Techniques of Validation

- Unreplicated Systems – Observed vs Predicted Regression
  - Important Assumptions of Linear Regression:
    - The X_i must be known exactly.
    - The variance of the errors must be constant for all values of X_i .
    - The Y_i are independent.

8.3 The Techniques of Validation

- Unreplicated Systems – Indices
  - Inequality coefficient (Theil's *U*) : Accurate models have small *U*.

$$U = \frac{\sqrt{\frac{1}{n}\sum (X_i - Y_i)^2}}{\sqrt{\frac{1}{n}\sum X_i^2} + \sqrt{\frac{1}{n}\sum Y_i^2}}$$

  - Mean square error of predictions, MSEP:

$$MSEP = \frac{1}{n}\sum (X_i - Y_i)^2 = (\overline{X} - \overline{Y})^2 + (S_X - rS_Y)^2 + (1 - r^2)S_Y^2 \qquad (8.3)$$

8.3 The Techniques of Validation

Unreplicated Systems – Indices

- MSEP is composed of three components associated with:
  1. Differences between the model and system means (i.e., a nonzero intercept or bias error): MC,
  2. Differences between the variance of model output and the variance of observations (i.e., slope-not-unity error): SC,
  3. The deviation of the correlation of model and observation values from 1.0 (i.e., random error): RC.

$$1 = \mathrm{MC} + \mathrm{SC} + \mathrm{RC}$$

$$= \frac{(\overline{X} - \overline{Y})^2}{MSEP} + \frac{(S_X - rS_Y)^2}{MSEP} + \frac{(1 - r^2)S_Y^2}{MSEP}$$

8.3 The Techniques of Validation

Unreplicated Systems – Indices

- A number of additional indices can be defined to further quantify model. To a certain extent, these indices can be thought of as *measures of model adequacy.*
- Mean absolute error, MAE:

$$MAE = \frac{\sum |Y_i - X_i|}{n}$$

- Mean absolute percent error, MA%E:

$$MA\%E = \frac{100}{n} \sum \frac{|Y_i - X_i|}{|Y_i|}$$

8.3 The Techniques of Validation

Unreplicated Systems – Indices

- Two indices that scale the model error to the variability of the observed data are *model efficiency* (EF) and the *Janus coefficient* (J^2)

$$EF = 1 - \frac{\sum_{i=1}^{m}(Y_i - X_i)^2}{\sum_{i=1}^{n}(Y_i - \overline{Y})^2}$$

$$J^2 = \frac{\sum_{i=1}^{m}(Y_i - X_i)^2 / m}{\sum_{i=1}^{n}(Y_i' - X_i')^2 / n}$$

where Y_i' and X_i' are the data and model predictions, respectively, for the dataset used to create and parameterize the model, and where *m* and *n* are the sample sizes of the two comparisons.

8.3 The Techniques of Validation

Unreplicated Systems – Indices

- Power (1993) defined *model accuracy* as $n(1 - EF)$ and suggests an F test of the hypothesis that accuracy equals 0 (m and $n - g$ degrees of freedom, g=number of parameters).
- Power also calls the numerator of J^2 the model's *predictive error* and the denominator the model's *replicative error.*
- With a few exceptions, these indices do not have inferential capabilities, but can be used to measure the degree of departure of model output from observations.

8.3 The Techniques of Validation

- **Unreplicated Systems – Multiple Responses**
  - All previous methods presume a single response variable, but models with several state variables are typical. One solution is to repeat the analyses for each response independently. Alternatively, one can analyze indices as the sum over all response variables:

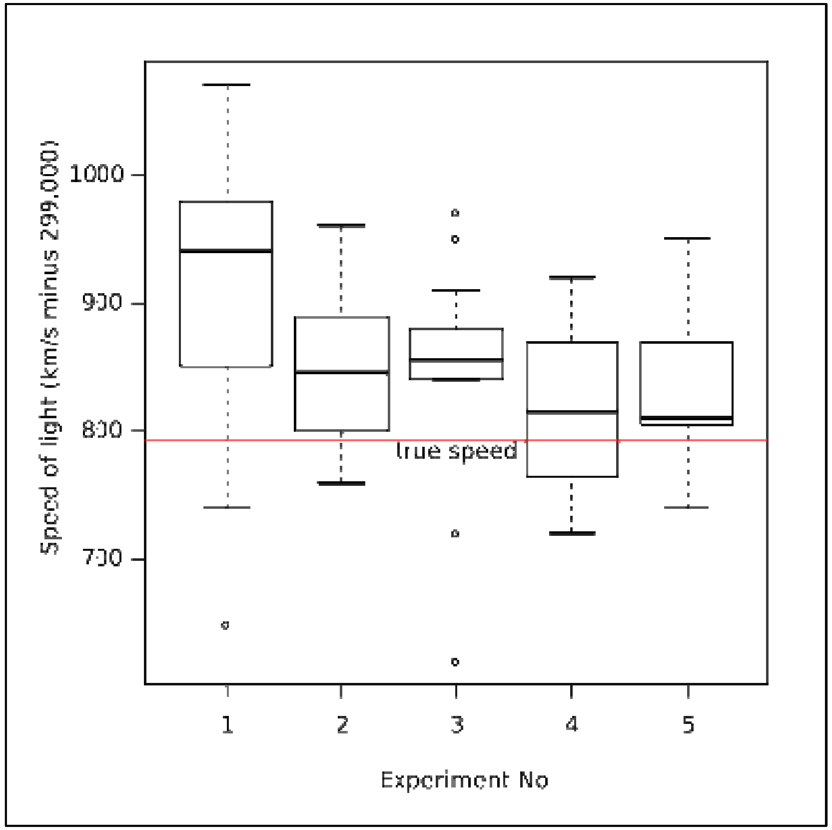
$$U_S = \frac{1}{K}\sum_{k=1}^{K} U_k \quad \text{or} \quad MSEP_s = \sum_{k=1}^{K} MSEP_k \qquad (8.4)$$

8.3 The Techniques of Validation

- Replicated Systems or Models
  - Replication in the system observations means that we have multiple, independent observations at points in time.
  - Model replication means we have a stochastic model that has been run several times or a deterministic model that is run several times with randomly selected parameter values.
  - An excellent tool for visualizing model or data variability, especially stochastic model output, is the boxplot. This is a graphical representation of a set of numbers in which the sample size, mean, median, the range, and other measures are all represented.

8.3 The Techniques of Validation

◎ Replicated Systems or Models - Boxplot



8.3 The Techniques of Validation

- Replicated Systems or Models – Single Value
  - If only a single value (e.g., the maximum of a state variable) is being tested, then standard statistical testing can be done (e.g., *t*-tests or ANOVA).
  - If variability exists in only one component (e.g., the data) then we use a single-sample *t*-test (H_o: $\mu_M = \mu_D$) . This compares the mean of the replicated data with the single number of the unreplicated number (model prediction).
  - If both model and data are variable, the two-sample *t*-test is used.

8.3 The Techniques of Validation

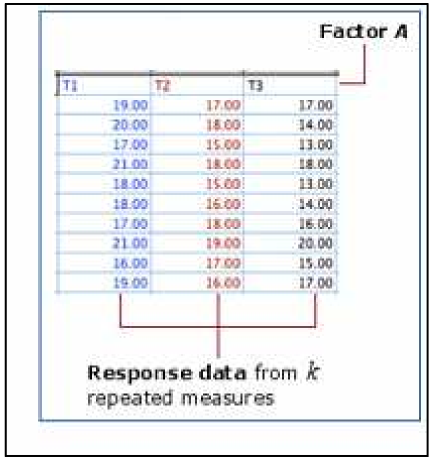
- Replicated Systems or Models – Time Series
  - As with unreplicated situations, time series introduce autocorrelations.
  - Measured values in real systems also tend to be correlated. These correlations can violate basic assumptions of standard statistical analyses so that extreme care must be exercised in their applications.
  - *Single-factor repeated measures analyses* and *split-plot designs* or *the multivariate profile analysis* are more appropriate techniques for time series analysis.

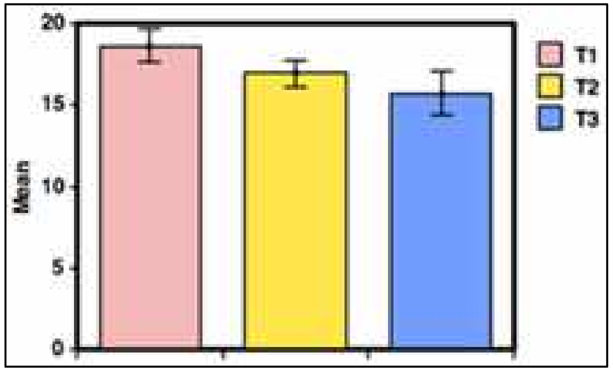
8.3 The Techniques of Validation

- Replicated Systems or Models – Time Series
  - Repeated Measures
    - Single-factor repeated measures and split-plot designs are types of analysis of variance (ANOVA).
    - Single-factor repeated measures designs use a single set of treatments applied sequentially to all of a single group of individuals (e.g., a sequence of drugs applied to patients).
    - A split-plot design applied to repeated measures situations generalizes this approach to include multiple factors so that not all individuals receive all treatments (e.g., drugs partitioned by chemical properties).
    - A split-plot design partitions the error among a main effect (e.g., system or location) and subdivides or splits each of these error components into effects associated with the treatments.
    - Both approaches assume that the correlation of responses among treatments is known and is constant over time.

8.3 THE TECHNIQUES OF VALIDATION

- Replicated Systems or Models – Time Series
  - Repeated Measures – Single-Factor Repeated Measures

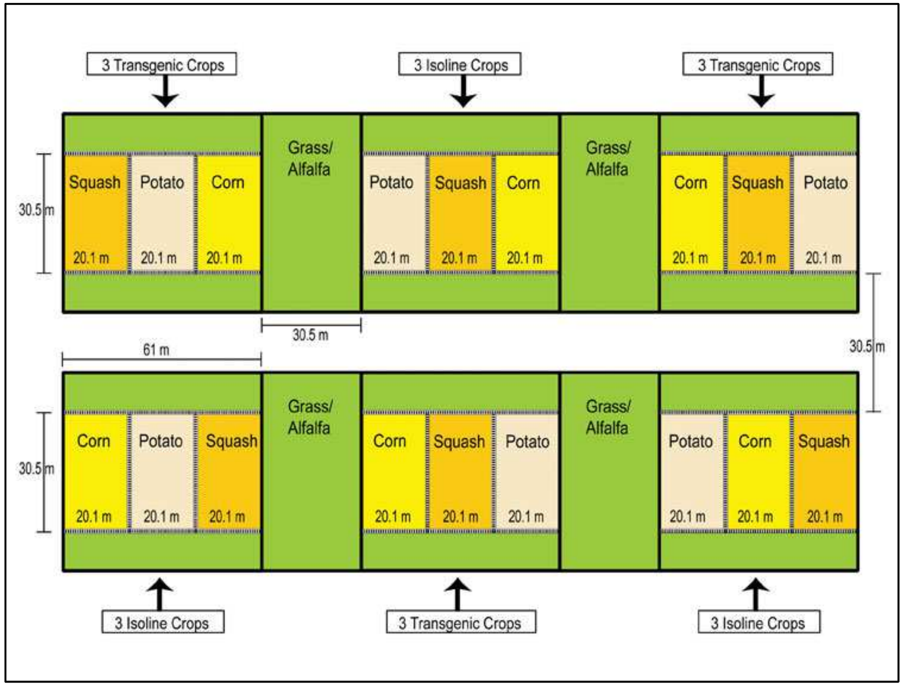




Single-Factor Repeated Measures ANOVA						
Source	*Sum of Squares*	*df*	*Mean Square*	*F*	*p*	*Partial* ω^2
Between Subjects	67.2	9	7.46667			
Time	42.4667	2	21.2333	15.7934	< 0.001	0.496533
Residual	24.2	18	1.34444			
Total	133.867	29				

8.3 The Techniques of Validation

- Replicated Systems or Models – Time Series
  - Repeated Measures – Split-plot Design



Split-plot experimental design consisting of 3 replicates of two genotypes: transgenic and isoline. Main plots were split among 3 crops consisting of sweet corn, potatoes, and winter squash. Transgenic cultivars targeted Lepidoptera in corn, Coleoptera in potato and aphid-transmitted viruses in squash. (Hoheisel and Fleischer, 2007)

8.3 The Techniques of Validation

- Replicated Systems or Models – Time Series
  - Profile Analysis
    - *Profile analysis* is a multivariate method that tests the hypothesis that the trajectories of data and model output are parallel.
    - The approach does not make assumptions about the nature of the variance or covariance relationships of the variables, so it is a more general approach to repeated measures problems.
    - Profile analysis permits us to examine the relation of the data and the model for several output variables (i.e., several state variables) simultaneously.

8.3 The Techniques of Validation

- Replicated Systems or Models – Time Series
  - Profile Analysis
    - For profile analysis, the null hypothesis tested is that the difference between model and data is 0.0 for each and all time values of comparison. This is analogous to the paired *t*-test.
    - Profile analysis calculates Hotelling's T^2 statistic, for which probability tables are available.

- Replicated Systems or Models – Time Series
  - Profile Analysis – Example
    - Data and Null Hypothesis:

$$H_0 : \mathbf{d}(1) - \mathbf{m}(1) = \mathbf{d}(2) - \mathbf{m}(2) = \cdots \mathbf{d}(k) - \mathbf{m}(k) = 0$$

Table 8.2: Hypothetical data and model response for six replicates and three points in time for phytoplankton (μg chl-a/liter) and zooplankton biomass (μg/liter). Columns are time (1,2,3); rows are the replicates and the model prediction.

	Phytoplankton			Zooplankton		
Sample	1	2	3	1	2	3
1	2.5	4.0	1.0	10	50	20
2	2.0	3.9	1.3	15	60	18
3	2.3	3.8	0.9	12	55	22
4	1.9	4.1	1.2	9	48	19
5	1.5	3.2	0.7	18	60	18
6	2.2	3.8	1.1	16	64	21
Model	2.1	3.8	1.0	13	56	20

- Replicated Systems or Models – Time Series
  - Profile Analysis – Example
    - Prediction Deviation:

Table 8.3: Deviations of data and model response for six replicates and three points in time for phytoplankton and zooplankton biomass. Columns are time, rows are replicates.

	Phytoplankton			Zooplankton		
Sample	1	2	3	1	2	3
1	0.4	0.2	0.0	–3	–6	0
2	–0.1	0.1	0.3	2	4	–2
3	0.2	0.0	–0.1	–1	–1	2
4	–0.2	0.3	0.2	–4	–8	–1
5	–0.6	–0.6	–0.3	5	4	–2
6	0.1	0.0	0.1	3	8	1

- Replicated Systems or Models – Time Series
  - Profile Analysis – Example
    - Time Differences:

Table 8.4: Time differences of model-data deviations for 6 replicates of phytoplankton and zooplankton responses. Columns are differences, rows are replicates. The dot in the column label (e.g., $\Delta_{P.1'}$) denotes all of the replicates in a given column. Column means are shown in the last row.

Sample	$\Delta_{P.1'}$ $\delta_{P\cdot1}-\delta_{P\cdot2}$	$\Delta_{P.2'}$ $\delta_{P\cdot2}-\delta_{P\cdot3}$	$\Delta_{Z.1'}$ $\delta_{Z\cdot1}-\delta_{Z\cdot2}$	$\Delta_{Z.2'}$ $\delta_{Z\cdot2}-\delta_{Z\cdot3}$
1	0.2	0.2	3	-6
2	–0.2	–0.2	–2	6
3	0.2	0.1	0	–3
4	–0.5	0.1	4	–7
5	–0.0	–0.3	1	6
6	0.1	–0.1	–5	7
Means →	–0.03	–0.03	0.17	0.50

8.3 THE TECHNIQUES OF VALIDATION

- Replicated Systems or Models – Time Series
  - Profile Analysis – Example
    - Table 8.4 is a one-sample multivariate test of the equality of means, and so is a generalization of the one-sample univariate test based on Student's *t*.
    - The test in the general case is based on Hotelling's T^2 for which the general formula for data of this type is

$$T^2 = (n)(\mathbf{Y} - Y_0)'\mathbf{S}^{-1}(\mathbf{Y} - Y_0)$$

where $\mathbf{Y}$-Y_0 is a column vector of the average differences between observed ($\mathbf{Y}$) and expected (Y_0) means, and $\mathbf{S}^{-1}$ is the inverse of the covariance matrix for the test variables

8.3 The Techniques of Validation

- Replicated Systems or Models – Time Series
  - Profile Analysis – Example
    - In this case, we are using the deviation of the model from the data; thus, the expected mean is 0, so Hotelling's T^2 for model validation is:

$$T^2 = (n)\mathbf{Y}'\mathbf{S}^{-1}\mathbf{Y} \qquad (8.5)$$

    - The variance-covariance matrix computed from Table 8.4 is

$$\mathbf{S} = \begin{bmatrix} 0.0747 & 0.0067 & -0.2933 & 0.2600 \\ 0.0067 & 0.0387 & 0.3267 & -1.1600 \\ -0.2933 & 0.3267 & 10.9667 & -17.5000 \\ 0.2600 & -1.1600 & -17.5000 & 42.7000 \end{bmatrix}$$

- Replicated Systems or Models – Time Series
  - Profile Analysis – Example
    - Provided that sufficient replicates are available, the inverse of S can be obtained from standard software packages as the matrix in:

$$T^2 = 6\left[\left[-0.03,-0.03,0.17,0.50\right]\begin{bmatrix} 30.44 & -147.43 & -4.28 & -5.94 \\ -147.43 & 1469.80 & 50.36 & 61.47 \\ -4.28 & 50.36 & 2.03 & 2.22 \\ -5.94 & 61.47 & 2.22 & 2.64 \end{bmatrix}\begin{bmatrix} -0.03 \\ -0.03 \\ 0.17 \\ 0.50 \end{bmatrix}\right]$$

$$= 0.0463$$

8.3 The Techniques of Validation

- Replicated Systems or Models – Time Series
  - Profile Analysis – Example
    - To determine the significance level of T^2, we use a table of Upper Percentage Points of Hotelling's T^2 for $T^{\alpha}(p, v)$, where p is $q(k-1)$ [i.e., 2(3 - 1) = 4], (α is the probability level for the test, and v is n - 1 (i.e., 5). The values for our case corresponding to α = 0.01, 0.05, and 0.10 are

$$T^{0.01}(4,5) = 992.494$$
$$T^{0.05}(4,5) = 192.468$$
$$T^{0.10}(4,5) = 92.434$$

    - Therefore, since 0.0463 < 992.494, we cannot reject the null hypothesis that the profile of data minus model predictions is zero at P = 0.01. Thus, this test result validates (or confirms) the model.

- Replicated Systems or Models – Time Series
  - Cross-correlation:
    - Qualitatively, this procedure attempts to quantify the correlation between two autocorrelated time series for a given lag. The lag accounts for the autocorrelation.
    - This method has the reputation of requiring large data sets. This requirement may limit its application in the ecological and environmental disciplines, but may not be a problem in biochemical and physiological systems.

8.3 The Techniques of Validation

- Replicated Systems or Models – Time Series
  - 95% Confidence Interval Method:
    - One can simply plot model output and the data on the same graph and count the number of times the model output (or mean model output) falls within the data's 95% confidence intervals. These intervals are

$$\overline{X} \pm [t_{(0.05,n-1)}]s_{\overline{X}}$$

    - This criterion is a possible measure of model adequacy.

Phytoplankton	Zooplankton
t_1 : 1.17 – 2.97	t_1 : 4.18 – 22.48
t_2 : 2.99 – 5.97	t_2 : 40.04 – 72.30
t_3 : 0.48 – 1.59	t_3 : 15.47 – 23.87.

8.4 Model Discrimination

- General Description
  - Model discrimination is an approach to content ourselves with deciding among a set of models based on their *relative adequacy*.
  - Calculating probabilities is central to model discrimination.
  - There are two broad types of model discrimination: *parametric* and *structural*.
  - Structural model discrimination is more closely allied with model validation.
  - There are three major, related approaches: *ratios of likelihood functions, information-based optimization criteria*, and *Bayesian inference.*

8.4 Model Discrimination

- Likelihood Functions
  - The *likelihood* of a sample is the probability that the sample would be drawn from a specified probability distribution with known parameters.
  - A likelihood function that calculates the likelihood of a sample is a mathematical function that results from applying a probability distribution to a particular sample in which one or more of the distribution parameters are allowed to vary as the function's independent variable.
  - The dependent variable of the likelihood function is the *a posteriori* probability of the sample given the underlying probability distribution.

8.4 Model Discrimination

Likelihood Functions – Is the Die True?

- For example, suppose we roll the die five times and observe two occurrences of the number 3. How likely is this outcome if the die is true?
- The underlying probability distribution for this kind of problem is the binomial distribution:

$$b(x;n,\theta) = \frac{n!}{x!(n-x)!}\theta^x(1-\theta)^{n-x}$$

- In the problem, however, we do not know the true θ, so we form the likelihood function:

$$L(\theta\,|\,[x,n]) = \frac{5!}{2!(3)!}\theta^2(1-\theta)^3 \qquad (8.6)$$

8.4 Model Discrimination

- Likelihood Functions – Is the Die True?

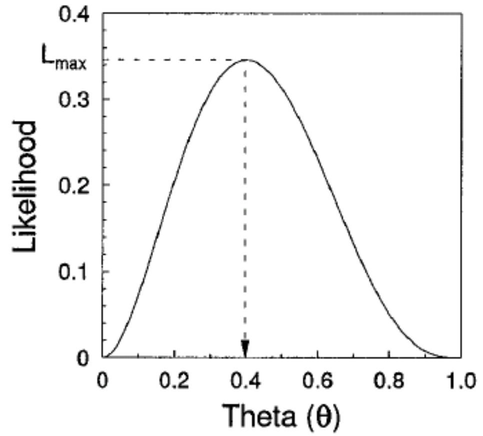


Figure 8.6: The likelihood function for the binomial probability distribution with $n = 5$ and $x = 2$. The maximum likelihood estimator is the θ associated with maximum of the function.

- From this we see that the probability of a face appearing that is associated with the maximum likelihood of the sample is 0.4, not 0.1667, which we would expect if the die were true. So this discrepancy between expected and most likely θ suggests that the die is not true.

8.4 Model Discrimination

- Likelihood Functions – Is the Die True?
  - We quantify the amount of discrepancy by forming the likelihood ratio (*R*): $L(\theta_{0.4}) / L(\theta_{0.17})$. In this case, the ratio is 2.15. So we say that the observed sample is 2.15 times as likely if $\theta = 0.4$ than if $\theta = 0.167$.
  - A rule of thumb, Reilly (1970) states that if R is greater than 10, then we consider the discrepancy is large enough to reject the hypothesis that the die is true.
  - Under certain conditions, the *log of the likelihood ratio* (log *R*) is distributed approximately as a χ^2 distribution, so that a probability can be associated with an observed *R* to assess if it is large enough to be due to factors other than chance.

8.4 Model Discrimination

- Likelihood Functions – Aside on Terminology?
  - In informal presentations, one often sees the likelihood function portrayed as:

    L(data I hypothesis)   or   *L*(data I model)

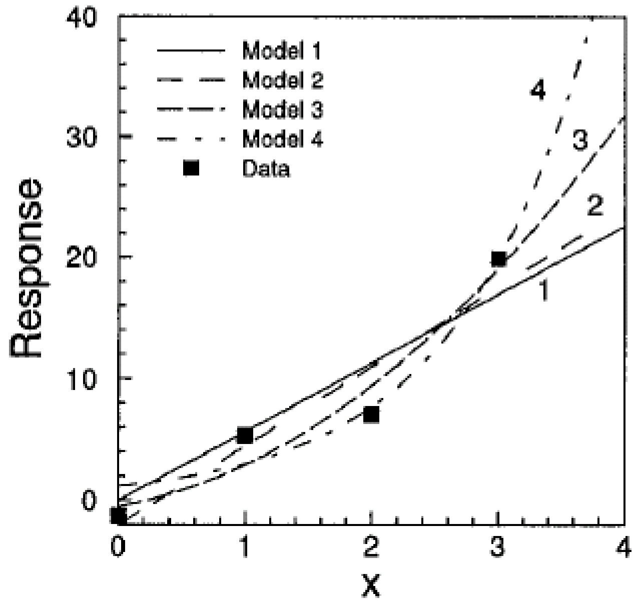
  - To compute L, we need not only the data *n* and *x* (Eq. 8.6), but also the value of θ, another argument of the function. So, the more correct presentation would be:

    L(θ I [model,**data**]),

  - *L* is the likelihood of observing dependent data as determined by variable parameters (θ) and independently observed data.

8.4 Model Discrimination

- Empirical Model Likelihood



x	0	1	2	3
y	-1.290	5.318	7.049	19.886

Model 1: $y = a_1 x$

Model 2: $y = a_0 + a_1 x$

Model 3: $y = a_0 + a_1 x + a_2 x^2$

Model 4: $y = a_3 e^{a_4 x}$

Figure 8.7: Four models fit to hypothetical data as a basis for discriminating among them. Model 1 is nested in Model 2 which is nested in Model 3.

8.4 Model Discrimination

Empirical Model Likelihood

- To compute the maximum likelihood for all four models, we need: (1) parameters to maximize, (2) some data, and (3) a probability distribution that depends on the model parameters.
- We also need a probability distribution that will compute the probability of observing the data, given a model.
- Each observed *y* value can be considered as being equal to a function plus an error term:

$$y_i = f(x_i, \theta_j) + \varepsilon_{ij} \qquad (8.7)$$

8.4 Model Discrimination

Empirical Model Likelihood

- The probability of the total error is just what we mean by the probability of observing the y_i. This, then, is the probability distribution we need for the likelihood of all the y. So, a general likelihood function is

$$L(\theta \mid \text{model}, \text{data}) = \prod_{i=1}^{n} p_i \qquad (8.8)$$

- The probability density function (pdf) for a single-variate normal distribution is

$$n(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\left(\frac{(x-\mu)^2}{2\sigma^2} \right) \right) \qquad (8.9)$$

Empirical Model Likelihood

- The particular likelihood function assuming normally distributed errors for all datum points (all y_i), a particular model j, and the independent data x needed by the model is

$$L_j(\theta\sigma^2 \,|\, [y_i, f_i(), x_i]) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\left(y_i - f(x_i, \theta_j)\right)^2}{2\sigma^2} \right)$$

$$= \left[\frac{1}{\sqrt{2\pi\sigma^2}} \right]^n \exp\left(-\frac{\sum_i^n \left(y_i - f(x_i, \theta_j)\right)^2}{2\sigma^2} \right) \qquad (8.10)$$

- This equation has the following important properties. (1) $\Sigma(y_i - f(x_i, \theta_j))^2$=RSS (residual sum of squares) is the least-squared error between data and model. (2) Models and parameter sets that have large errors (poor fits) have small likelihood values. (3) There is a single maximum, the maximum likelihood, which corresponds to the minimum of RSS, (4) The set of (θ_j, σ^2) associated with the maximum is the best set of model parameters for model i. These (θ_j, σ^2) are the *maximum likelihood estimator*s of the parameters.

8.4 Model Discrimination

- Empirical Model Likelihood
  - To fairly compare and discriminate among a set of models, we want to use, for each model, the model's parameters that make the data the most likely, i.e., the maximum likelihood estimates of θ_j and σ^2 for each model. When we have these estimates, we will also have the maximum likelihood estimate RSS/n = MSEP. When MSEP is estimated, and substituted for σ^2 in Eq. 8.10, the maximum likelihood function for model *j* is:

$$L_j(\theta\sigma^2 \mid [y_i, f_j(), x_i]) = \left[\frac{1}{2\pi\hat{\sigma}^2}\right]^{n/2} \exp(-\frac{n}{2}) \qquad (8.11)$$

$$\ln(L_j(\theta\sigma^2 \mid [y_i, f_j(), x_i])) = -\frac{n}{2}\ln(\hat{\sigma}^2) - \frac{n}{2}\ln(2\pi) - \frac{n}{2} \qquad (8.12)$$

8.4 Model Discrimination

Empirical Model Likelihood

- In the special case that the models are nested we can use the likelihood ratio test (LRT). Model A is nested in (simpler than) Model B if the former can be obtained from the latter by setting one or more parameters to zero.
- The likelihood ratio test is based on the ratio of likelihoods or, equivalently, the difference between the log-likelihood of the simpler model ($\ln(L_s)$) and the more complex model ($\ln(L_c)$). This quantity is χ^2 distributed with the null hypothesis that $\ln(L_s) = \ln(L_c)$ and tested with degrees of freedom equal to the difference in the number of parameters between the two models:

$$\chi^2 = -2[\ln(L_s) - \ln(L_c)]$$

Empirical Model Likelihood

Table 8.5: Likelihood comparisons of four models on data in Fig. 8.7

Model	RSS	$\hat{\sigma}^2$	$\ln(L_j)$	$\ln(L_i) - \ln(L_{\max=3})$	χ^2	df	P
1	28.465	7.116	–3.925	–1.603	3.206	2	0.201
2	22.473	5.618	–3.452	–1.130	2.260	1	0.133
3	12.773	3.193	–2.322	0.000	—	—	—
4	11.854	2.964	–2.173	—	—	—	—

8.4 Model Discrimination

Mechanistic Model Discrimination

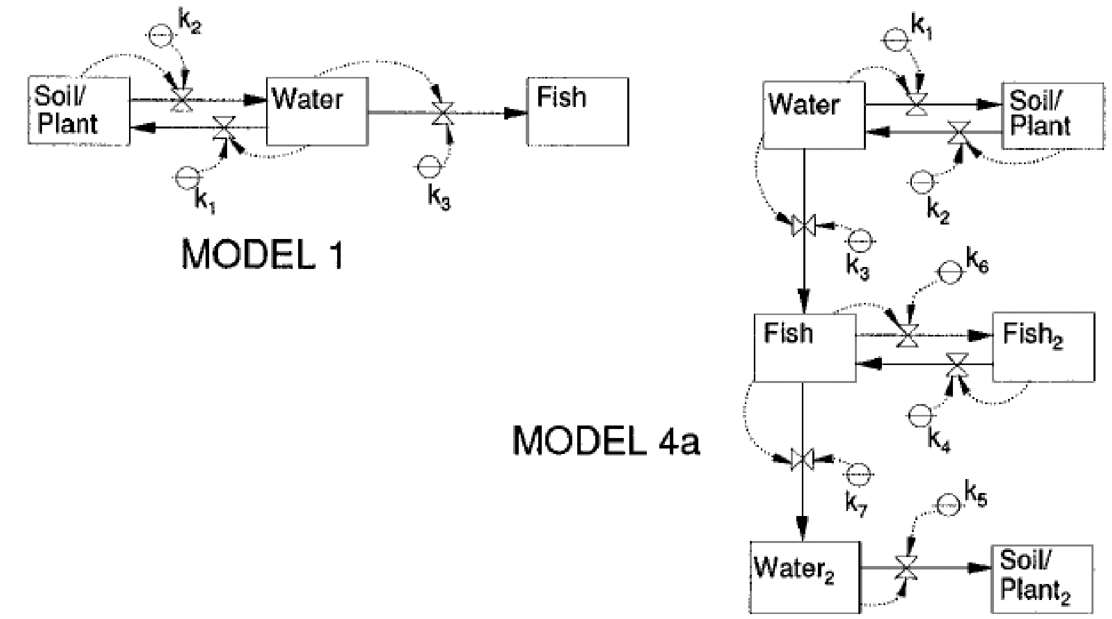


Figure 8.8: Two of the seven models of Dursban movement in an aquatic microcosm (Blau and Neely 1975). In model 4a, $Fish_2$, $Water_2$, and Soil/$Plant_2$ represent additional storage compartments for ^{14}C.

8.4 Model Discrimination

◎ Mechanistic Model Discrimination

Table 8.6: Maximum likelihood values, ratios, and chi-square values for the seven models of Dursban movement. $n = 36$. Not all models are nested in the best model (4a). df = degrees of freedom for the chi-square test = difference in number of parameters. P is the probability that a χ^2 value as large or larger than observed would occur by random sampling. AIC is the Akaike Information Criterion for each model; Δ_i is the difference between a model's AIC and the smallest AIC in the set of models.

Model	RSS	$\hat{\sigma}^2$	$\ln(L_j)$	$\ln(L_i/L_{\max=4a})$	χ^2	df	P	AIC	Δ_i
1	5374	149.3	–90.11	–81.34	162.7	4	$\ll 0.001$	188.22	154.7
2a	1964	54.56	–71.99	–63.22	—	—	—	153.98	120.44
2b	848	23.56	–56.87	–48.10	96.20	3	$\ll 0.001$	123.74	90.20
3a	208.3	5.786	–31.60	–22.83	45.66	2	$\ll 0.001$	75.20	41.66
3b	207.9	5.775	–31.56	–22.79	—	—	—	77.12	43.56
4a	58.6	1.628	–8.772	0.00	0.00	—	—	33.54	0.000
4b	79.4	2.206	–14.24	–5.468	—	—	—	42.48	8.940

8.4 Model Discrimination

- Information-based Discrimination
  - More parameters often produce functions with more complicated structure (e.g., curvi-linearity, or many maxima and minima), which might be better able to match complicated, non-smooth data.
  - However, all parameters are estimated with errors, and it often happens that the total error of the function is positively related to the number of parameters as they each contribute their individual errors. This is *error propagation* and is discussed in Chapter 9.
  - A number of schemes have been proposed to incorporate the number of parameters into the model discrimination process. All of these decrease model "quality" as the number of parameters increase.

8.4 Model Discrimination

- Information-based Discrimination
  - As Burnham and Anderson describe, if one model is considered the focal or "true" model, then a second model is viewed as approximating the first. The distance between the two is the information that is lost, if we use the second model in lieu of the true model.
  - Information theory provides a formula for the distance (the Kullback-Liebler distance) from a candidate model to a focal model based on particular values of the parameters required by the two models.

8.4 Model Discrimination

Information-based Discrimination - AIC

- The Akaike Information Criterion (AIC) provides an unbiased approximation that can be applied to empirical data based on the log-likelihood function. This approximation is computed for each model j:

$$AIC_j = -2\ln(L(\theta\hat{\sigma}^2 \,|\, [y, f_j(), x])) + 2K \qquad (8.13)$$

where *K* is the number of parameters estimated in fitting the model to the data. *K* equals all the unknown coefficients in the model itself plus parameters of the error distribution that must be specified. Thus, *K* represents a penalty we incur by using complicated models to represent the data.

8.4 Model Discrimination

- **Information-based Discrimination - AIC**
  - When the models analyzed by AIC are nested, there is a relationship between AIC and LRT:

$$LRT = AIC_l - AIC_j + 2k \qquad (8.14)$$

  - Burnham and Anderson (1998) suggest, as a rough rule of thumb, that if $\Delta_l \leq 2$ ($\Delta_l = AIC_l - AIC_{min}$), model *l* performs similarly to the best model and should not be eliminated; if $\Delta_l > 10$, model *l* is not close in quality to the best model and can be eliminated.
  - One major failing of this approach is the absence of a hypothesis test. A different data set might (and often will) suggest a different model as being best. One obvious solution to this sampling distribution problem is to collect many datasets and calculate an AIC for each dataset and model.

8.4 Model Discrimination

Bayesian Inference

- The likelihood method quantitatively ranks the adequacy of a set of competing models by their ability to fit the data, but it does not actually compute the probability that the models are correct. One method of calculating this probability uses *Bayes' Theorem*.
- Bayesian statistics are based on a different set of probabilities, and, in particular, include estimates of the truth of the null hypothesis prior to the test being made. Thus, they permit the inclusion of prior knowledge (e.g., data from other similar systems, historical data, expert opinion, etc.) in the test of the hypothesis for the given data set.

8.4 Model Discrimination

- Bayesian Inference
  - The basis for this approach to inference is Bayes' Theorem which in the present context is a recipe for calculating the probability that model is true, given the observed data and a finite set of m alternative models. The Bayesian probability is

$$P(M_i \mid \mathrm{Y}) = \frac{P(M_i)P(\mathrm{Y} \mid M_i)}{\sum_{j=1}^{m} P(M_j)P(\mathrm{Y} \mid M_j)} \qquad (8.15)$$

where m is the number of alternative models, P(M_i) is the *prior* probability that model *i* is true, and P(**Y** I M_i) is the probability of observing **Y** values given that M_i is true. This latter quantity is typically estimated as the *maximum likelihood estimator* of **Y**. The denominator is a scaling factor that normalizes the likelihood of a particular model to the total likelihood of all the models.

8.4 Model Discrimination

- Bayesian Inference
  - Other users of Bayesian inference, however, recommend not using any previous experience. They suggest assigning the prior of each model an equal probability: $1/m$, where m is the number of models. Such priors are termed *noninformative*.
  - The likelihood functions computed using the optimal fit of parameters to the data can be used as the P(**Y** I M_i) in Bayes' Theorem.

$$P(M_i \mid \mathrm{Y}) = \frac{P(M_i) L(\theta_i \hat{\sigma}^2 \mid [y, M_i, x])}{\sum_{j=1}^{m} P(M_j) L(\theta_j \hat{\sigma}^2 \mid [y, M_j, x])} \qquad (8.16)$$

8.4 Model Discrimination

- Bayesian Inference

Table 8.7: Bayesian posterior probabilities of seven Dursban models. Column 2 = prior probability of model i, column 3 = Likelihood of model, column 4 = posterior probability of model i. (Recalculated from Blau and Neely (1975) and Carpenter (1990).)

Model	$P(M_i)$	L_i	$P(M_i \mid \mathbf{Y})$
1	0.1429	1.049×10^{-40}	4.716×10^{-36}
2a	0.1429	7.763×10^{-33}	3.491×10^{-28}
2b	0.1429	2.861×10^{-26}	1.287×10^{-21}
3a	0.1429	2.700×10^{-15}	1.214×10^{-10}
3b	0.1429	2.809×10^{-15}	1.263×10^{-10}
$\boxed{\text{4a}}$	0.1429	2.214×10^{-5}	$\boxed{0.9958}$
4b	0.1429	9.344×10^{-8}	4.202×10^{-3}
Denominator = 2.224×10^{-5}			

8.5 Meta-Models

- A meta-model is a nonlinear regression model of the output of a dynamic model.
- The method developed by Kleijnen is as follows.
  1. Use a series of original model runs to generate a data set.
  2. Identify a set of potentially interesting relationships. Then, fit linear or nonlinear functions to the model data set.
  3. Validate the meta-model by running the original model a second set of times with different input values (e.g., different driving temperatures). If valid, the meta-model should correctly predict the quantitative meta-relationships of the new runs.
- A valid meta-model will characterize the important dynamic relationships that are produced by the mechanistic relationships used in the original model. The meta-model can then be further validated against empirical data.

8.6 Précis on Validation

- The relation of model validation and model discrimination has yet to be firmly established. They share important statistical similarities, and combined with carefully designed independent experiments.
- They represent different philosophies toward model evaluation.
- In practice, statistical validation emphasizes model adequacy; incorporating model complexity into our ultimate assessments of model performance may be one approach to measuring model reliability.
- The issue of subjectivity in all aspects of model evaluation, whether it comes from model choice or prior probabilities, will continue to be hotly debated for many years.